SAIDS — PYQ Question Mapper

Statistics for Artificial Intelligence & Data Science

CSE-AIML Sem V · C Scheme (R-2019) · Dec 2023 – May 2025

PAPER LEGEND

Dec 2022	Dec 2023	May 2024	Dec 2024	May 2025 (AIML)	May 2025 (IT)	IAE QB-1	IAE QB-2

QUESTIONS PER MODULE (ALL PYQs)

Module I	Statistical Foundations, Covariance & Sampling	43 Qs
Module II	Statistical Inference, Estimation & Resampling	45 Qs
Module III	Regression Models, Regularization & Nonlinear Models	13 Qs
Module IV	Classification & Supervised Learning	7 Qs
Module V	Unsupervised Learning & Dimensionality Reduction	8 Qs
Module VI	Model Evaluation, Bayesian Inference & Ethics	3 Qs

Module I Statistical Foundations, Covariance & Sampling

Syllabus Topics: *Data types (categorical, numerical, ordinal) · Descriptive stats (central tendency, dispersion, skewness, kurtosis) · Data viz (histograms, boxplots, scatter plots, pair plots, QQ plots) · Probability distributions (Normal, Binomial, Poisson, Exponential, Uniform, Beta, Gamma, Weibull) · Covariance, correlation, correlation matrices · Sampling (simple random, stratified, cluster) · Central Limit Theorem (CLT) & Law of Large Numbers · Empirical CDF · Data preprocessing (outlier treatment, encoding, normalization)*

Total: 43 questions · Dec 2022: 7 · Dec 2023: 6 · May 2024: 7 · Dec 2024: 8 · May 2025 (AIML): 6 · May 2025 (IT): 5 · IAE QB-1: 4

Paper	Q No.	Question / Topic	Type
Dec 2022	Q1a	Find standard deviation of average temperatures over 5-day period: 19, 21, 18, 24, 12	*Numerical*
Dec 2022	Q1b	Normal distribution (μ=30, σ=4): find P(x<40) and P(30<x<35)	*Numerical*
Dec 2022	Q1e	What do you mean by correlation and regression? Explain with example	*Theory*
Dec 2022	Q2a	Find correlation coefficient from Age (X) vs Glucose Level (Y) data (6 subjects)	*Numerical*
Dec 2022	Q3a	What is the use of scatter plot and box plot?	*Theory*
Dec 2022	Q6a	Binomial distribution: insurance agent, P(living 30+ yrs)=2/3 for 5 people — P(all 5), P(≥3), P(exactly 2)	*Numerical*
Dec 2022	Q6b	Short notes (any two): Central Limit Theorem, Standard Error	*Short notes*
Dec 2023	Q1d	Short note on Random sampling	*Short note*
Dec 2023	Q1e	What is the Empirical CDF function?	*Short note*
Dec 2023	Q2a	Frequency distribution table for 30 orange weights; find class mark, range, number of intervals, lowest-frequency interval	*Numerical*
Dec 2023	Q3b	Concept of correlation; how is it different from regression?	*Theory*
Dec 2023	Q4a	Normal distribution: P(car speed > 100 km/hr) with μ=90, σ=10	*Numerical*
Dec 2023	Q4b	Explain Numerical and Categorical data types with appropriate examples	*Theory*

Paper	Q No.	Question / Topic	Type
May 2024	Q1d	What do you mean by correlation and regression? Explain with example	*Theory*
May 2024	Q2a	Normal distribution (μ=30, σ=4): find P(x<40), P(x>21), P(30<x<35)	*Numerical*
May 2024	Q2b	Poisson distribution: vehicles at 300/hr — P(none in 1 min), expected in 2 min, P(expected number passes in 2 min)	*Numerical*
May 2024	Q3a	Normal distribution: P(battery charge lasts 50–70 hrs) for μ=50, σ=15	*Numerical*
May 2024	Q4a	Explain QQ plots in detail; show how scatterplots explore relationships between variables	*Theory*
May 2024	Q6a	Explain Binomial distribution in detail; Type I error concept in bottle volume test	*Theory + Numerical*
May 2024	Q6b	Short notes (any 2): Chi-square distribution, Weibull distribution, Stem & Leaf Plot, Box Plot	*Short notes*
Dec 2024	Q1a	Explain percentiles and Boxplots with example	*Short note*
Dec 2024	Q1b	Illustrate Central Limit Theorem with a neat diagram	*Short note*
Dec 2024	Q1d	What is a Histogram? Give its applications	*Short note*
Dec 2024	Q2b	Explain Normal and Poisson Distribution; GMAT scores (μ=527, σ=112): P(score>500) and score for top 5%	*Theory + Numerical*
Dec 2024	Q4b	Mean, SD, standard error and Confidence Interval; compute all for a 10-data-point sample	*Theory + Numerical*
Dec 2024	Q5b	Continuous Probability Distribution & PDF; Uniform distribution on [0,20]: P(5≤X≤15), P(X<10), P(X>18)	*Numerical*
Dec 2024	Q6b	Short note: Histogram and Scatterplot with example	*Short note*
Dec 2024	Q6d	Short note: Weibull Distribution	*Short note*
May 2025 (AIML)	Q1c	Explain the difference between Stratified and Cluster Sampling	*Short note*
May 2025 (AIML)	Q1e	Define standard deviation and interquartile range with examples	*Short note*
May 2025 (AIML)	Q2a	Find the correlation coefficient from given Experience (X) vs Salary (Y) data	*Numerical*
May 2025 (AIML)	Q4a	Z-score: sample mean=250, population mean=260, SD=40; calculate z-score for sample mean	*Numerical*
May 2025 (AIML)	Q4b	Frequency distribution table for 50 college students' weekly study hours	*Numerical*
May 2025 (AIML)	Q6b	Short notes (any 2): Chi-square distribution, Weibull distribution, Stem & Leaf Plot, Box Plot	*Short notes*
May 2025 (IT)	Q1b	Explain Poisson probability distribution with one example	*Short note*
May 2025 (IT)	Q1d	Discuss Quantitative Data, Descriptive Statistics, Variables, Categorical data	*Theory*
May 2025 (IT)	Q1e	Why is sampling necessary? Describe population and sample mean	*Short note*
May 2025 (IT)	Q2b	Binomial distribution: insurance agent, P(living 30+ yrs)=2/3 for 5 people — P(all 5), P(≥3), P(exactly 2)	*Numerical*
May 2025 (IT)	Q6b	Short notes: Scatter diagram, Sample space, Discrete probability distribution	*Short notes*
IAE QB-1	A2	What is mean? Find the mean of the data: 4, 6, 8, 10	*Numerical*
IAE QB-1	B1	Find the mean, median and mode for the following marks: 10, 20, 20, 30, 40	*Numerical*
IAE QB-1	B5	A university has 2000 students. Explain an appropriate sampling technique to select a sample; use CLT to explain how statistical inference can be drawn from the sample	*Theory*
IAE QB-1	B6	Binomial distribution: A factory produces bulbs with P(defective)=0.1.	*Numerical*

Paper	Q No.	Question / Topic	Type
		If n=5 bulbs are selected, find P(exactly 2 are defective)	

Module II Statistical Inference, Estimation & Resampling

Syllabus Topics: *Point estimates · Bias, Variance, MSE · MLE & Method of Moments · Confidence intervals (mean, proportion, variance) · Hypothesis testing: z-test, t-test, Chi-square, F-test, ANOVA (1-way, 2-way) · Nonparametric tests: Mann-Whitney U, Kruskal-Wallis, Friedman, Wilcoxon, Sign test · Bootstrap CI & resampling · Likelihood Ratio Tests (LRT) · Type I & II errors · p-value*

Total: 45 questions · Dec 2022: 8 · Dec 2023: 7 · May 2024: 7 · Dec 2024: 8 · May 2025 (AIML): 7 · May 2025 (IT): 5 · IAE QB-1: 3

Paper	Q No.	Question / Topic	Type
Dec 2022	Q1c	Discuss Bootstrapping vs. re-sampling	*Short note*
Dec 2022	Q1d	School principal: state null and alternative hypotheses (teachers claim students use computer 3.2 hrs/day)	*Theory*
Dec 2022	Q2b	Explain why ANOVA is used; solve One-way ANOVA for Observations A, B, C (5×3 table)	*Theory + Numerical*
Dec 2022	Q3a	Explain Type I & Type II error in detail	*Theory*
Dec 2022	Q3b	Kruskal-Wallis test: 4 manufacturing facilities, exam scores after machining training, at 95% confidence — are scores same?	*Numerical*
Dec 2022	Q4a	t-score: sample mean=290, expected mean=300, SD=50, n=15 — find t-score	*Numerical*
Dec 2022	Q5a	z-test: pea plant pods (μ=145, σ=100, $\bar{x}$=147) — state hypotheses, test statistic at α=0.05	*Numerical*
Dec 2022	Q6b	Short note: Confidence Interval	*Short note*
Dec 2023	Q1a	Short note on hypothesis testing	*Short note*
Dec 2023	Q1b	What is Fisher's exact test?	*Short note*
Dec 2023	Q2b	Purpose of one-way ANOVA; key components: dependent variable, independent variable, factors	*Theory*
Dec 2023	Q3a	Standard error of estimate for household children data; 95% confidence interval	*Numerical*
Dec 2023	Q5a	t-test: Duracell CEO claims μ=300h; sample n=20, $\bar{x}$=270h, s=50h — reject claim?	*Numerical*
Dec 2023	Q6a	z-test: pea plant pods (μ=145, σ=100, sample $\bar{x}$=147) — state hypotheses and test statistic	*Numerical*
Dec 2023	Q6b	What is Chi-Square Test and in what situations is it commonly used?	*Theory*
May 2024	Q1a	Define Confidence Interval	*Short note*
May 2024	Q1b	Standard Error of sample mean: μ=$12, σ=$3, n=30 — estimate and interpret SE	*Numerical*
May 2024	Q1c	What is hypothesis testing? Explain Type I and Type II errors	*Short note*
May 2024	Q1e	What is analysis of variance (ANOVA)? Explain its usage	*Short note*
May 2024	Q3b	z-test: score change (μ=80, σ=10, n=38, $\bar{x}$=88) at α=0.05 — evidence for claim?	*Numerical*
May 2024	Q4b	One-way ANOVA numerical: four samples A, B, C, D — identify differences	*Numerical*
May 2024	Q5a	F-Test: F=2.38, df=(8,3) — find F-table value; reject null at α=0.05 (one-tailed)?	*Numerical*
Dec 2024	Q1c	Explain Null and alternative Hypothesis with example	*Short note*
Dec 2024	Q1f	What is the F-Test used for?	*Short note*
Dec 2024	Q2a	Chi-Square Test: gender vs. coffee/tea preference contingency table (Male/Female × Coffee/Tea) at α=0.05	*Numerical*
Dec 2024	Q3b	t-Distribution: CEO claims bulb lasts 300 days; n=15, $\bar{x}$=290, s=50 — P($\bar{x}$ ≤ 290)	*Numerical*
Dec 2024	Q4a	ANOVA: why use it; one-way vs two-way difference; solve one-way	*Theory +*

Paper	Q No.	Question / Topic	Type
		ANOVA for 3 groups	*Numerical*
Dec 2024	Q5a	Nonparametric vs parametric methods; Friedman Test: THM water cleanup data (12 counties)	*Theory + Numerical*
Dec 2024	Q6a	Short note: Type I and Type II errors in detail	*Short note*
Dec 2024	Q6c	Short note: Bootstrapping vs Re-sampling	*Short note*
May 2025 (AIML)	Q1a	What is hypothesis testing? Explain Type I and Type II errors	*Short note*
May 2025 (AIML)	Q1b	What is Fisher's exact test?	*Short note*
May 2025 (AIML)	Q2b	Chi-Square Test: gender vs. online/in-store shopping (n=200) at α=0.05	*Numerical*
May 2025 (AIML)	Q3a	Explain the concept of p-value in hypothesis testing	*Theory*
May 2025 (AIML)	Q3b	Kruskal-Wallis test: aptitude scores of Grade A, B, C students at 95% confidence level	*Numerical*
May 2025 (AIML)	Q5a	z-test: pharmaceutical blood pressure drug (μ=10mmHg, σ=5, n=30, $\bar{x}$=12) at α=0.05 — state H_0/H_1, test stat, conclusion	*Numerical*
May 2025 (AIML)	Q6a	Two-way ANOVA: concept; difference from one-way ANOVA; assumptions; Friedman's test as nonparametric alternative	*Theory*
May 2025 (IT)	Q1c	What is statistical inference?	*Short note*
May 2025 (IT)	Q2a	Null and alternate hypothesis; explain Type I and Type II error	*Theory*
May 2025 (IT)	Q5a	t-test: pizza shop cheese weights (n=7) — state hypotheses, test statistic; reject at 10%, 5%, 1% level?	*Numerical*
May 2025 (IT)	Q6a	What is Sign Test? Explain Wilcoxon Signed-Rank Test	*Theory*
May 2025 (IT)	Q6b	Short note: Kruskal-Wallis Test	*Short note*
IAE QB-1	A7	What is bias of an estimator? Explain its significance	*Theory*
IAE QB-1	A8	What is Maximum Likelihood Estimation (MLE)?	*Theory*
IAE QB-1	B3	z-test: A manufacturer claims batteries last μ=50h. Test at 5% significance: n=36, $\bar{x}$=53h, σ=9h — state H_0/H_1, compute test statistic, and give conclusion	*Numerical*

Module III Regression Models, Regularization & Nonlinear Models

Syllabus Topics: *Simple & multiple linear regression · LLSR derivation · Model assumptions · Residual analysis · Multicollinearity (VIF) · Polynomial & interaction effects · Nonlinear regression · Regularization: Ridge, Lasso, Elastic Net · Spline regression & LOESS · GLM: Logistic regression (binary), Poisson (count data) · Model evaluation: MAE, MSE, RMSE, R^2, Adjusted R^2*

Total: 13 questions • Dec 2022: 2 · Dec 2023: 2 · May 2024: 1 · Dec 2024: 1 · May 2025 (AIML): 2 · May 2025 (IT): 4 · IAE QB-1: 1

Paper	Q No.	Question / Topic	Type
Dec 2022	Q4b	Multiple regression: find relation between GPA of students and number of study hours and height	*Numerical*
Dec 2022	Q5b	Simple linear regression equation + coefficient of determination for Hour–Temperature data	*Numerical*
Dec 2023	Q1c	Short note on Simple Linear Regression	*Short note*
Dec 2023	Q5b	Explain Linear Least Square Regression (LLSR) with advantages and disadvantages	*Theory*
May 2024	Q5b	Find simple linear regression equation and coefficient of determination (R^2) for given X–Y data	*Numerical*
Dec 2024	Q3a	Explain single and Multiple linear regression with example and suitable plot	*Theory*
May 2025 (AIML)	Q1d	Explain Linear Regression and its Applications	*Short note*
May 2025 (AIML)	Q5b	Find the simple linear regression equation for Time–Growth data	*Numerical*
May 2025 (IT)	Q3b	Differentiate between Simple Linear Regression and Multiple Regression	*Theory*
May 2025 (IT)	Q4a	What is Logistic Regression?	*Theory*
May 2025 (IT)	Q4b	Find simple linear regression equation for Bill–Tip data	*Numerical*
May 2025 (IT)	Q5b	Derive Simple Linear Regression parameters to predict the value of the output	*Derivation*
IAE QB-1	A10	What is multicollinearity in regression? Explain its effect on the model	*Theory*

Module IV Classification & Supervised Learning

Syllabus Topics: *Classification metrics (accuracy, precision, recall, F1, ROC-AUC, PR curves) · Logistic regression, odds ratio · Naive Bayes · k-NN classifier · Decision Trees & pruning · Ensemble methods: Bagging, Random Forest, Gradient Boosting · SVM (linear & kernel) · Handling imbalanced data: SMOTE · Confusion matrix analysis*

Total: 7 questions · IAE QB-2: 7

Paper	Q No.	Question / Topic	Type
IAE QB-2	A1	What is the difference between precision and recall? Mention one situation where recall is preferred over precision	*Theory*
IAE QB-2	A2	A classification model produces TP=40, FP=10, FN=5, TN=45. Calculate the precision of the model	*Numerical*
IAE QB-2	A3	What is SMOTE? Why is it used when handling an imbalanced classification dataset?	*Theory*
IAE QB-2	A4	What is the kernel trick in SVM? State its purpose	*Theory*
IAE QB-2	B1	Classification model for fraudulent transactions (imbalanced dataset): a) Why can accuracy be misleading? b) How do precision, recall, and F1-score provide better information? c) Which metric to prioritize to detect as many fraudulent transactions as possible?	*Theory*
IAE QB-2	B2	A Decision Tree gives 95% training accuracy but 78% testing accuracy. a) Identify the likely problem. b) Explain how pruning can help. c) State one advantage of Random Forest over a single Decision Tree	*Theory*
IAE QB-2	B3	Explain the working principle of SVM: a) What is a hyperplane? b) What is the margin and why is maximizing it important? c) Explain the kernel trick with one suitable example	*Theory*

💡 **Note:** No questions from PYQs map to this module. All 7 questions above are from the IAE Question Bank (QB-2) only. Classification algorithms remain high-priority for upcoming exams.

Module V Unsupervised Learning & Dimensionality Reduction

Syllabus Topics: *PCA (covariance matrix, Eigen decomposition, explained variance) · LDA (supervised dim. reduction) · Kernel PCA & ICA (conceptual) · Clustering: K-means, Hierarchical, DBSCAN · Cluster evaluation: silhouette score, Davies-Bouldin index · Dimensionality reduction: t-SNE, UMAP · Outlier detection: Z-score, IQR, Isolation Forest, LOF*

Total: 8 questions · Dec 2024: 1 · IAE QB-2: 7

Paper	Q No.	Question / Topic	Type
Dec 2024	Q1e	How to detect outliers?	*Short note*
IAE QB-2	A5	What is the purpose of explained variance in PCA? How does it help in selecting the number of principal components?	*Theory*
IAE QB-2	A6	Mean of feature=50, SD=5. Calculate Z-score for observation value=65. Is it an outlier using $\|Z\|>3$ criterion?	*Numerical*
IAE QB-2	A7	What is the main difference between PCA and LDA in terms of whether they use class labels?	*Theory*
IAE QB-2	A8	What is the silhouette score in clustering? What does a high silhouette score indicate about cluster quality?	*Theory*
IAE QB-2	B4	PCA on correlated features: a) Explain how PCA uses covariance matrix and eigen decomposition to reduce dimensionality. b) What is explained variance? c) Why is retaining the first few principal components useful for ML models?	*Theory*
IAE QB-2	B5	Customer transaction dataset: a) Explain K-means clustering for identifying customer groups. b) State one limitation of K-means and how DBSCAN addresses it. c) Explain how silhouette score evaluates cluster quality	*Theory*
IAE QB-2	B6	High-dimensional data with potential outliers: a) Explain purpose of t-SNE or UMAP for visualizing high-dimensional data. b) Explain IQR method for outlier detection. c) Name one tree-based method for outlier detection and briefly explain its idea	*Theory*

💡 **Note:** Only 1 question from PYQs (outlier detection). 7 additional questions are from the IAE Question Bank (QB-2). PCA, K-means, DBSCAN, t-SNE, and silhouette score are high-potential exam topics.

Module VI Model Evaluation, Bayesian Inference & Ethics

Syllabus Topics: *Model selection: AIC, BIC · Bias-variance trade-off · Cross-validation: k-fold, LOOCV · Calibration curves & Brier score · Bayesian fundamentals: prior, likelihood, posterior · MAP vs MLE · Conjugate priors · Naive Bayes from Bayesian view · Interpretability: SHAP, LIME, PDP, ICE · Fairness in ML · Ethical considerations · Data privacy · Reproducible ML (MLflow, W&B)*

Total: 3 questions · May 2025 (IT): 1 · IAE QB-2: 2

Paper	Q No.	Question / Topic	Type
May 2025 (IT)	Q1a	What do you mean by random variables? Explain Bayes theorem	*Theory*
IAE QB-2	A9	Differentiate between AIC and BIC as model selection criteria. Mention one key difference in how they penalize model complexity	*Theory*
IAE QB-2	A10	A model has high training accuracy but significantly lower validation accuracy. Identify the likely issue using the bias–variance trade-off	*Theory*

💡 **Note:** Only 1 question from PYQs (Bayes theorem). 2 additional questions are from the IAE Question Bank (QB-2). AIC/BIC and bias-variance trade-off are now covered via IAE QB. Cross-validation, SHAP/LIME, and ML fairness remain untested.

KEY TAKEAWAYS FOR EXAM PREP

1. **Modules I & II dominate every paper (~65–70% of marks)**
   Every paper has hypothesis testing questions (z-test, t-test, ANOVA, Chi-square), normal/Poisson distribution numericals, confidence intervals, and descriptive statistics. Prioritise these two modules above all others.
2. **Module III (Regression) appears in every paper**
   Expect at least 1–2 numerical (find the SLR equation for given data) and 1 theory question (explain LLSR / multiple regression). Know how to compute regression coefficients, R^2, and residuals.
3. **Hypothesis testing numericals are repeated every exam**
   z-test, t-test, one-way ANOVA, and Chi-square independence tests have numerical problems in every paper. Memorise the formulas: test statistic, degrees of freedom, and decision rule.
4. **Nonparametric tests are increasing (Dec 2024 → May 2025)**
   Friedman test, Kruskal-Wallis test, and Wilcoxon Signed-Rank test appeared in the newer papers. Prepare these as short-note or theory questions, not just numericals.
5. **Modules IV, V, VI: IAE QB covers what PYQs miss**
   Classification (SVM, Decision Trees, Random Forest, SMOTE), PCA/clustering (K-means, DBSCAN, silhouette score), and model evaluation (AIC, BIC, bias-variance) are covered in the IAE Question Bank but absent from PYQs. Prepare SVM working principle, PCA with eigen decomposition, and confusion-matrix numericals as priority topics.
6. **Recurring short-note topics across papers**
   Histogram, Box Plot, Stem & Leaf, Weibull distribution, Type I/II errors, Fisher's exact test, and Bootstrapping appear as short notes in multiple papers. Prepare 5-line answers for each.